12 — Liquid Track: Validated Numbers, Errata, and the Gating Test Plan

v1.4 — the quantitative record at DP-A

Two independent first-principles validation passes plus a cross-track resolution pass were run. Everything below is the surviving, corrected set — with the error trail kept deliberately visible.

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1. Validated quantities (DP-A: 32 °C / 80%, ω 24.2 g/kg, sink ~29 °C)

Quantity	Value	Derivation
Ambient ω	**24.2 g/kg**	Psat 4.76 kPa → Pv 3.81 → 0.622·3.81/97.5
Target ω, 29 °C / 55%	**13.8 g/kg**	Psat 4.01 → Pv 2.20
Brine floor, base 40 wt% (aw 0.45)	**11.9 g/kg at 30 °C brine** (10.0 @27 / 14.2 @33)	temperature-dependent band
Brine floor, hot-regen 43–44 wt% (aw 0.34)	**9.0 g/kg at 30 °C** (7.5 @27)	"
Fresh-air floor (4 occupants)	**48 m³/h ≈ 58 kg/h**	12 m³/h·person, CO_2-interlocked
ERV pre-dry (ε 0.8)	fresh 24.2 → **15.9 g/kg**	ω_amb – ε(ω_amb – ω_cab)
Mixed-mode total absorber flow	**~123 m³/h (≈147 kg/h)**	hold 13.8: gains 280 g/h ÷ (13.8 – 11.9) g/kg. One quantity in two units at ρ 1.2 (doc 00 §2), not a range
Peak removal, 4 adults mixed-mode	**0.88 kg/h** (0.49 with ε 0.8 ERV... see note)	CV: fresh import + occupant/envelope gains. Bare-fresh 0.88; ERV'd steady removal ~0.5–0.6
Peak regeneration heat	**~0.92 kW bare** / **~0.6 kW ERV'd**	removal × 2.44 MJ/kg ÷ COP. The 0.92 kW figure is at **COP 0.65** — the with-recovery-HX value two rows down, not the 0.59 no-recovery build-up
Daily heat, occupied DP-A duty	**17–20 kWh bare → ~9–11 kWh with ERV(ε 0.8) + DCV**	duty-scheduled
CO_2-battery heat (doc 00 §5)	**~1.6–2.5 kWh/day** for 1.6 kg CO_2	1.0–1.3 kWh/kg incl. water co-adsorption + sensible
COP build-up (no recovery HX)	**≈0.59**	latent 0.678 + brine sensible 0.184 + air sensible 0.131 kWh/kg, ×1.15 losses; recovery HX → ~0.65
Solar array (ERV'd budget)	**2.5–4.5 m²** flat-plate class / ~2 m² ETC top-lift with PVT preheat	9–11 kWh ÷ 2.5–3 kWh/m²·day
Diesel heater-only, full duty	**~1.1–1.4 L/day**	÷ 8.5 kWh/L
Absorption heat into brine	**0.66 kW peak**	0.88 kg/h × 2.7 MJ/kg
Uncooled contactor self-limits	+5–8 K → floor 12–14+ g/kg	air removes only ~16 W/K per 48 m³/h
Cooling leverage	**0.5–0.7 g/kg per °C** of brine cooling	aw 0.45 band; land cool-water sinks exploit this
Raw-water cooling flow	**~300 L/h** base (2 K rise) / 600–900 heat-rich	0.66 … 2 kW rejection
Unattended thresholds, 1 contactor	**~90 m³** holding 13 g/kg / **~160–215 m³** mold-safe	120 g/h capacity vs 0.1 ACH × Δω
Overnight reserve (4 adults)	**~35–40 kg** concentrate	12 h duty ÷ 0.143
CO_2 at the ventilation floor	1,920 awake / 1,400 asleep	0.072 m³/h ÷ 48 — **above the <1,000 spec; closed by the doc 00 §5 stack**
Berth cascade M-cycle	supply 22.6 °C · ~102 W/outlet · ~4.1 L/day	working ⅓ from cabin, exhaust sat ~27 °C
Whole-cabin once-through (X1, why deferred)	5.5–11 kg/h absorber duty at 1–2 kW sensible	370–750 m³/h all-ambient supply
Still driving force	10–65 kPa ΔPv	pool 60–93 °C vs 29 °C-sink condenser (4–5 kPa)
Still footprint, 0.9 kg/h	0.5–0.9 m² tray	~1–2 kg/h·m² at design ΔT
Air-swept exhaust (degraded)	~98 g/kg, dew point **53 °C** → condenser recovers ~64 g/kg	X8 rule 3; condensate technical-grade PENDING TDS
Crystallization liquidus	40 wt% ≈ **12–13 °C**; 42 ≈ 18–19; 44 ≈ 22 °C	$CaCl_2$ solubility curve
Electrical, film primary	**25–50 W** initial / **50–80 W** scaled (+ERV/exhaust fan)	fans 25–40 + recirc 15–30 + raw water 5–15 + peristaltic
Electrical, column annex	~100–140 W drilled rings; 170–380 W membrane diffusers — avoid	erratum 3

2. Errata — the correction trail (kept visible on purpose)

1. **Regeneration heat overstated 2×** in first-pass tables; corrected by re-derivation. *Lesson: never cite a margin as a computed value.*
2. **Absorber outlet is a band, not a point** (temperature-dependent); absorber cooling reclassified optional → **required**.
3. **Diffuser dynamic wet pressure was missing** from the column blower budget — membranes silently double–triple electrical draw; drilled rings win.
4. **Nylon removed** from approved materials ($CaCl_2$ ESC of polyamides).
5. **Crystallization threshold corrected** (40 wt% liquidus ~12–13 °C, not ~5); interlock restated at 42/43 wt%.
6. **Regen exhaust condensate**: 53 °C-dew-point exhaust rains salty condensate — sloped drip-leg routing mandatory; the sealed still is the logical endpoint, and the X8 condenser (doc 11 §4) is the corrected degraded mode.
7. **Sparger head constant** 1.35–1.4 kPa per 10 cm at SG 1.4.
8. **Once-through could not deliver 4-adult comfort at the governing point** (59–66% RH) — mixed-mode adopted as the occupied baseline;

once-through re-scoped to unattended/low-occupancy. *Lesson: re-run every comfort claim when the design point moves.*

9. **The self-consistent steady state, not the hand chain, sizes the system** — cabin humidity, supply state, and duty must be solved together (doc 00 §4); hand-chained state tables drifted 15–30% on duty.
10. **OPEN — the ERV effectiveness behind the ERV'd duty line.** Doc 10 §3 carries an ERV pre-dried fresh state of 17.4 g/kg; §1 above carries 15.9 g/kg. Only 15.9 follows from the specified ε ≥0.8 (24.2 − 0.8 × 10.4); 17.4 implies ε ≈ 0.65. The published ERV'd figures — 0.49 kg/h removal and ~0.6 kW — track 17.4, so they are effectively the ε 0.65 case wearing an ε 0.8 label; at a true ε 0.8 the removal falls to ~0.40 kg/h. Both values are left standing and flagged rather than reconciled on paper, because **test E measures the real ε_lat** (and its salt-aerosol fouling trend) and will settle which case the design actually operates in. Nothing downstream is load-bearing on the difference: the bare-fresh 0.88 kg/h peak governs the sizing. *Lesson: an effectiveness quoted beside a duty is a claim about both — check that the duty was computed at the effectiveness printed next to it.*

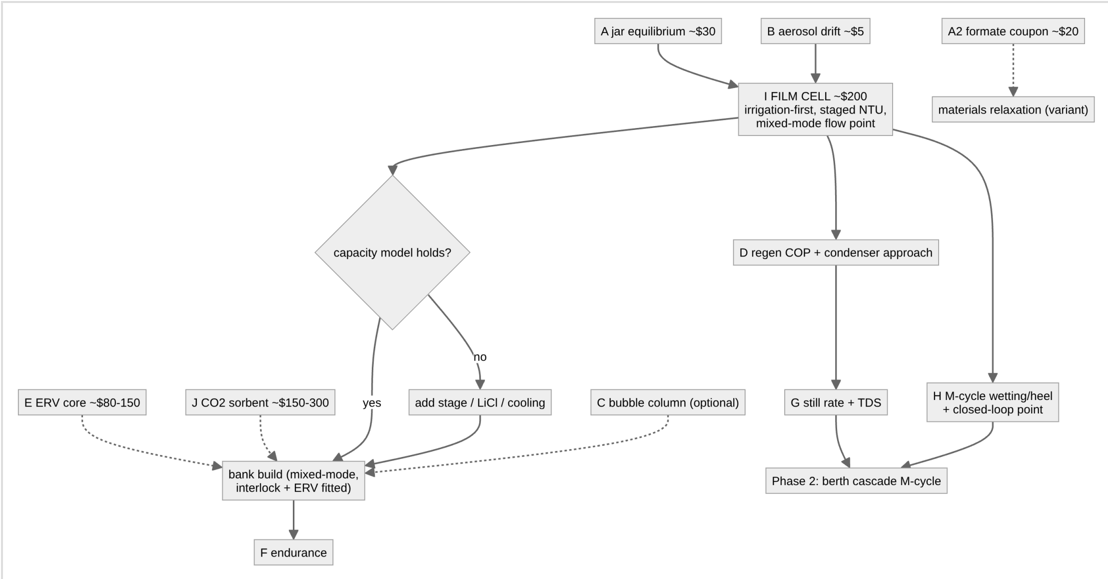
3. Dominant sensitivities

Parameter	Uncertainty	Effect
$CaCl_2$ water activity vs wt% and T	published data ±0.05 aw	±1.5–3 g/kg on the floor — the largest unknown. **Measure (test A)**
Film K·a with brine	literature 1–3 kg/m^3·s	contactor depth 1–3 stages (test I)
Wetting rate	150–240 L/min·m^2 demanded	underwetting collapses K·a nonlinearly — test I varies irrigation *first*
ERV real ε_lat + fouling	vendor data vs salt aerosol	gates the 9–11 kWh/day budget (test E)
CO_2-sorbent working capacity / **water co-adsorption** / **slip**	DAC literature vs this regime	gates X10 entirely (test J)
Occupant moisture generation	50–90 g/h·person	±30% on load
Infiltration (unattended)	0.05–0.15 ACH	one vs two contactors
Drift-eliminator + demister effectiveness	unquantified for brine	gates cabin connection (test B)
Fan/pump efficiencies	factor ~2 at small scale	measure real amps

4. Test plan — cheapest decisive experiment first

Test	Cost	Decides
A — jar equilibrium (days)	~$30	Real aw table: sealed jars, RH probe over $CaCl_2$ 35/38/40/42/44 wt% (+LiCl arm) at ~27 and ~33 °C. **Go/no-go: 40 wt% @ 33 °C ≤ 55% ERH**
A2 — aerated hot coupon (weeks)	~$20	Whether potassium-formate brine survives *aerated* 80 °C service — materials relaxation for that variant only
B — aerosol drift (1 week)	~$5	Bare mild-steel coupon downstream of eliminator + demister, max face velocity, trickle *and* flooded. Any rust = redesign before cabin connection. Also arbitrates the solid track's liquid-desiccant verdict (X3)
E — ERV core (weeks)	~$80–150	Real ε_lat; salt-aerosol fouling trend (U-tube ΔP); condensation at DP-A inlet; **CO_2 crossover** / **EATR <5%**
I — film cell prototype (weeks, **THE GATE**)	~$150–250	Outlet RH vs **irrigation rate** first; 1/2/3-stage NTU/m; face velocity incl. the **~123 m^3/h mixed-mode point**; rocking rig 5–20°; flooded run; wet/dry crystallization hygiene
J — CO_2 sorbent (weeks)	~$150–300	Working capacity at 1,000–1,500 ppm / 45–55% RH; desorption completeness 85–95 °C in the half-cycle; water co-adsorption penalty; **amine**/**ammonia slip (breathing-air gate)**; oxidative fade 10^2–10^3 cycles. Gates X10; fallback = oversized ERV+DCV
J-K — potash CO_2 sorbent (weeks; waste-heat platform variant)	~$40–80	K_2CO_3-on-apolar-carbon: capacity at 1,000–1,500 ppm / 45–55% RH; regeneration completeness + kinetics at 120/135/150 °C with logged purge; support screen (apolar carbon vs TiO_2) incl. an **alumina-deactivation control**; caking/deliquescence over wet–dry cycles; **alkaline particulate**/**mist carryover (breathing-air gate)**; 10^2-cycle stability. Runs only where a ≥ ~130 °C tap exists (X11)
L — hot-film absorption (weeks; upgrade path X12, doc 31 §2 — waste-heat platforms only)	~$60–120	Absorption rate and approach of a 43–44 wt% film at 85/88/90 °C against ~20–25 kPa steam; drain-back-on-stop and a deliberate cool-in-place fault. Reuses the sous-vide + wet/dry-bulb stack; test A's aw table feeds the lift-ceiling inequality directly. **Gates the AHT**
A3 — crystallizer jar extension (days; upgrade path, doc 31 §6 — mass-limited platforms only)	~$10	Seeded vs unseeded 44 wt% parked at ~25 °C: supercooling degree, phase formed, caking over ~50–100 park/dissolve cycles, redissolution rate. Shares test A's jars and instrumentation. **Gates the static crystallizer pot**
D — regeneration COP (days)	~$50	Still tray + air-swept at 60/70/85 °C incl. PVC-tray 60 °C point and **condenser approach**
G — sealed-still rate (days)	~$40	kg/h·m^2 vs pool temp; condensate TDS (entrainment ≈ 0); **extended: TDS on the air-swept condenser stream**
H — M-cycle wetting/heel (days)	~$60	Supply temp vs feed humidity; 5/10/15° tilt; **closed-loop feed point (X8: recycled dried exhaust)** — shared verbatim with the solid track
F — endurance (months, passive)	~$0	Locker/closet-dryer duty: salt creep, fouling, crystallization events, ΔP trend
C — bubble column (optional)	~$150	Only if the annex is ordered; sparger DWP shootout

On cost figures. The per-test estimates above are kept because they carry the argument that matters — these are jar-and-coupon experiments, not instrumented lab campaigns — and because they let a reader sequence the program by cost. A single aggregate headline has been **retired**: the former `~$485–755` did not decompose from this table, its span implied a costing exercise that was never done, and a currency-and-date-specific total is the first thing to rot in a record meant to be read decades from now. Read the column as **2026 order-of- magnitude, one currency, one region**. The claim the program actually rests on is scale, not precision: **the decisive liquid-track experiments together cost of order a few hundred dollars**, and test A alone — the one that moves the largest single unknown — is about $30.



5. Rejected CO_2-removal alternatives (quantified record; see doc 00 §5)

Zeolites (water outcompetes CO_2 at any cabin humidity) · seawater/raw-water absorption (>6 m^3/h fully equilibrated, slow hydration kinetics, re-humidifies the air) · membranes (0.1 kPa partial pressure) · algae/plants (150+ kWh/day of light for 3.4 kg/day) · liquid amines (volatile slip into breathing air) · soda lime/LiOH/KO_2 (~15 kg/day — emergency consumable only) · electrochemical capture (R&D watch item, ~1 kWh_e/kg in principle, not procurable at scale). Added by the X11 survey: open-metal-site MOFs (Mg-MOF-74 class — water poisons and hydrolyzes the sites, ~16% of capacity retained humid; poisoned-site recovery sits beyond bus grade) · flue-gas physisorbent MOFs (CALF-20 — ~0.1–0.17 mmol/g at cabin ppm, selectivity ceiling ~40–47% RH straddling the bed placement; MUF-16 class — humid-tolerant but flue-pressure-validated, marginal at 0.1–0.15 kPa, cobalt-dust gate) · fluorinated ultramicroporous physisorbents (SIFSIX/TIFSIX/NbOFFIVE — genuinely ppm-capable at ~1.2–1.3 mmol/g, but 7.5–10 mmol/g water co-adsorption rides every swing, SIFSIX-3-Ni phase-changes at ~50% RH, humid/oxidative cycling instability, and a fluorometallate-pillar degradation path upstream of breathing air) · moisture-swing quaternary-ammonium resins (commodity AERs, but regeneration-by-wetting evaporates its water into the cabin and re-imports it onto the brine loop — order 5–15 kWh/day of displaced regeneration heat vs 1.6–2.5 for TSA, an X8 rule-1 violation at DP-A; capture also degrades at moderate-plus RH) · supported K_2CO_3 **at solar grade** (equilibrium-dead below ~120 °C — adopted instead as the waste-heat-platform bed: X11, test J-K).

6. Open questions (logged, not load-bearing)

- Evaporative-media life in concentrated $CaCl_2$ through wet/dry cycles (inside I/F).
- LiCl sourcing vs accepting hot-regen $CaCl_2$ as the deep-dry lever; LiCl worsens wetting (revalidate irrigation margin if spiked).
- Fans/pumps: independent units vs single + manifold — measured amps settle it.
- Cool-humid (non-tropical) absorber behavior; cold-climate land installs.
- Heat-driven air pumps (NIFTE/fluidyne/thermoacoustic) — R&D track only.
- Upgrade-path watch items (doc 31, none load-bearing): AHT absorber materials at 90 °C brine (PVDF vs PP margin); MVR compressor mist carryover (test-B analogue pre-gate) and non-condensables purge cadence; coupled-HP refrigerant choice (R290 procurement vs R600a fit); hexahydrate PCM nucleator persistence (logged, low priority at DP-A).
- Distillate potability chain (baffle + TDS + carbon polish): one lab water test before regular drinking (safety register item 5).
- CO_2-sorbent watch register (research-grade, no supply chain; any promotion inherits full test-J scope): **MOF-808-AA** (glycine/lysine-appended Zr-MOF — nonvolatile anchored amines, water-*enhanced* uptake at ~50% RH, mild desorption) primary; **tetraamine-Mg$_2$(dobpdc)** (supersedes the mmen/diamine class, whose amine volatilization its own authors concede; steam-regenerable, ppm-validated) secondary; SIFSIX-18-Ni-β a literature line only (hydrophobic trace-CO_2 physisorbent — no supply chain, fluoride-pillar caveat).

7. Bottom line

The physics closes on independent passes and one cross-track resolution pass. The occupied claim now reads: **4-adult comfort at DP-A stands on the mixed-mode baseline with cooled base brine, ERV, and the CO_2 stack** — conditional on tests A/B/E/I/J, all bench-cheap. Two-adult and unattended claims stand on the base once-through system. Phase 2 (berth cascade M-cycle) adds no new load-bearing physics *in the cascade topology*; whole-cabin AC is deferred on heat-scale grounds, where the two tracks converge.

Version history

- **v1.0** — New lineage from archived core-03: all quantities restated at DP-A; errata 8–9 added from the resolution passes; tests E and J and the P16 extensions integrated; CO_2-alternatives record added.
- **v1.1** — Test J-K added (§4); CO_2-rejection record extended with the X11 survey — open-metal-site MOFs, CALF-20, MUF-16, fluorinated HUMs, moisture-swing AERs, solar-grade potash (§5); the mmen upgrade line replaced by the CO_2-sorbent watch register — MOF-808-AA primary, tetraamine-Mg$_2$(dobpdc) secondary (§6).
- **v1.2** — Upgrade-path tests L (AHT hot-film absorption, X12) and A3 (crystallizer jar extension) added to §4 as platform-conditional entries outside

the baseline gating budget; doc 31 watch items added to §6.

- **v1.4** — Aggregate cost headline retired from the §4 title (it did not decompose from the table and dates badly in a prior-art record); per-test estimates retained and explicitly scoped as 2026 order-of-magnitude figures.
- **v1.3** — Clarifying pass from the parameter register (doc 50 §3.5), no design figure changed: §1's absorber-flow row restated as one quantity in two units (~123 $m^3/h \approx 147$ kg/h) rather than a range, and the COP behind the 0.92 kW peak named as the with-recovery 0.65; **erratum 10 opened** — the ε_{lat} inconsistency behind the ERV'd duty line, left standing and gated on test E; §4 gains a budget note recording that the ~$485–755 headline no longer decomposes from the table beneath it.